# ER Diagram

The system as built on branch `mini-erp` (Version 2.0): the three changes (Purchase Logic Draft2, QT Upgrade 2.0, User Version 2.0) are in. This replaces the older single-version description.

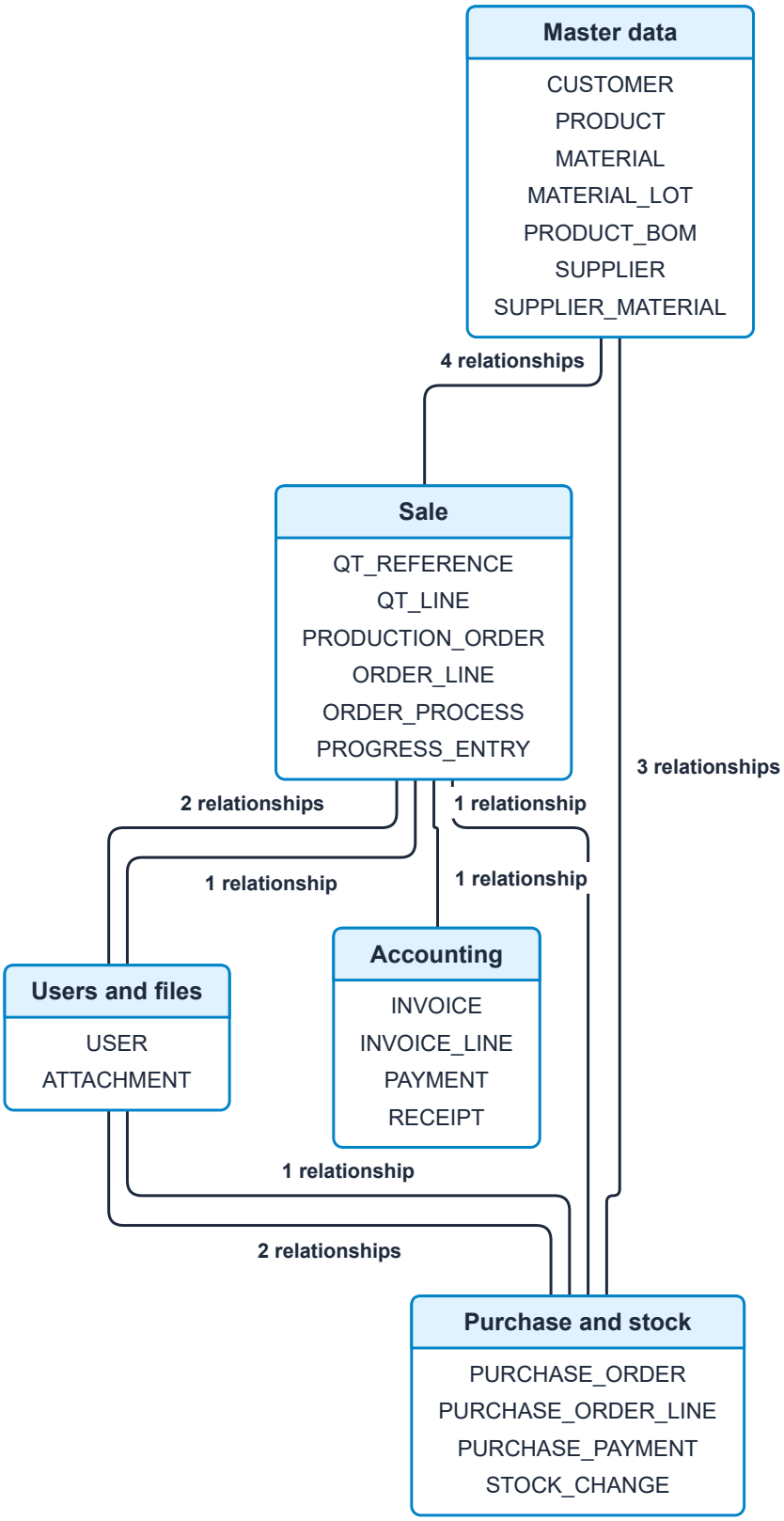



Figure 1. The 23 tables in 5 groups, and the relationships between the groups

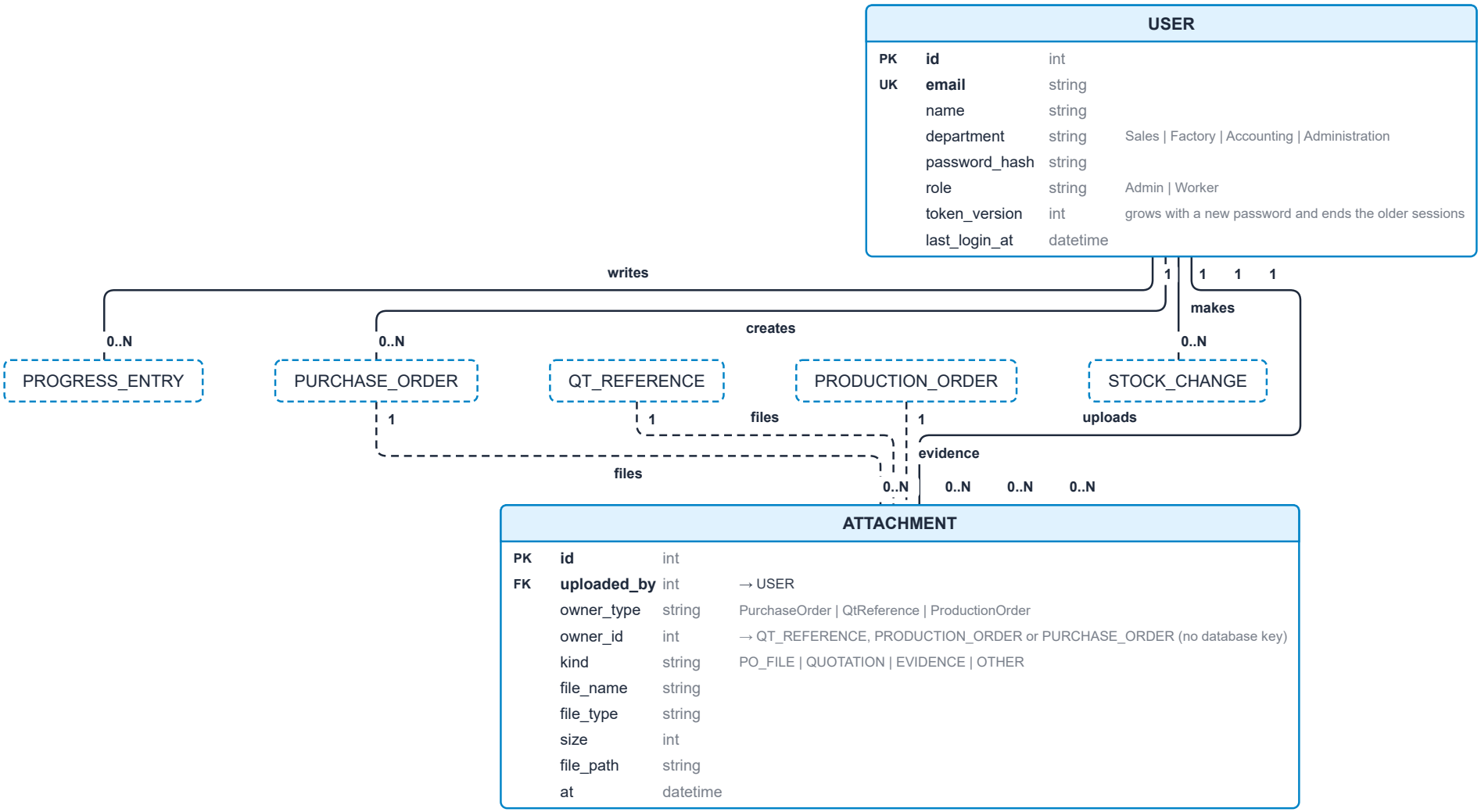



Figure 2. Users and files (1 = exactly one, 0..1 = none or one, 1..N = one or many, 0..N = none, one or many; → = the foreign key points to)

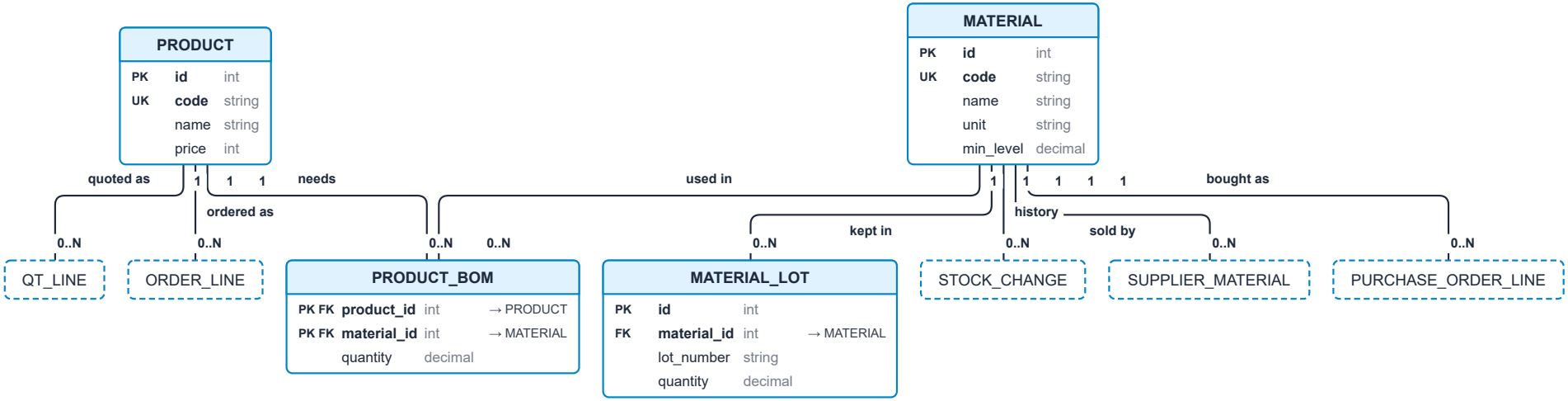



Figure 3. Master data: products and materials (1 = exactly one, 0..1 = none or one, 1..N = one or many, 0..N = none, one or many; → = the foreign key points to)

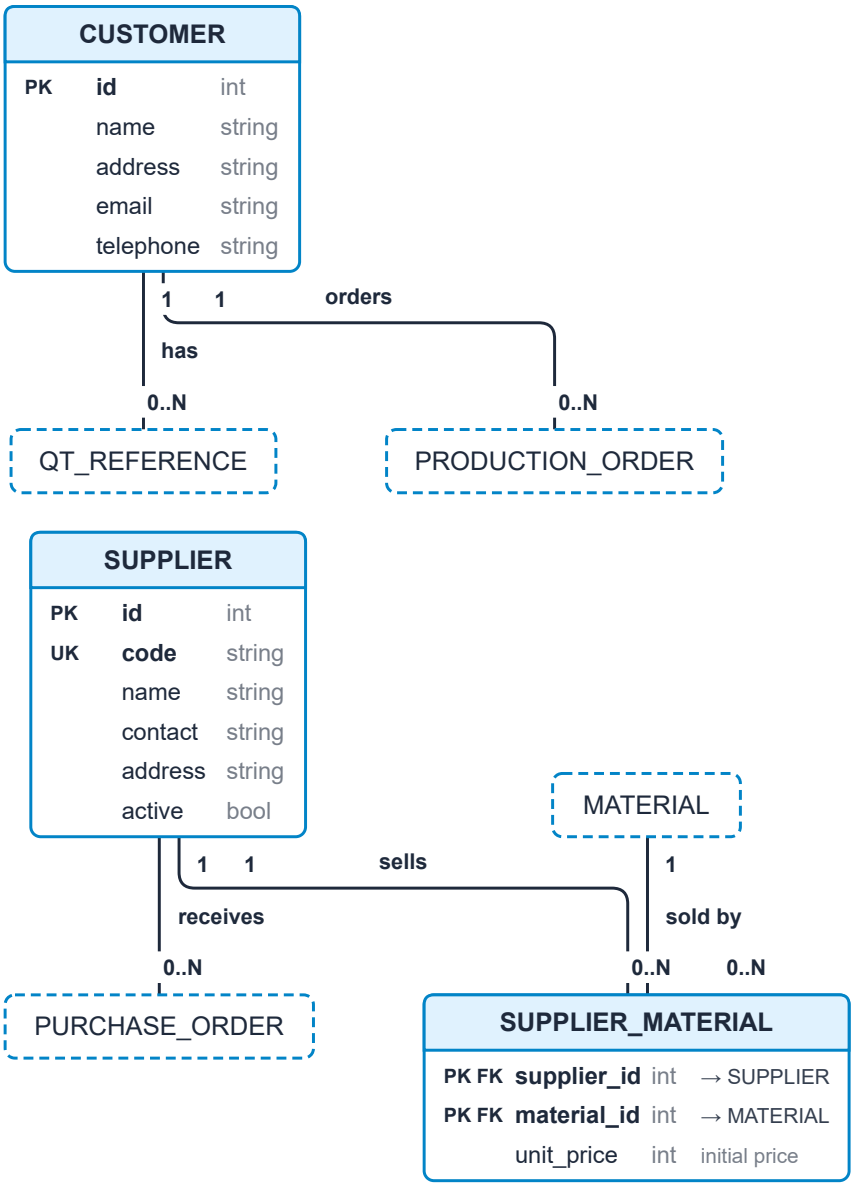



Figure 4. Master data: customers and suppliers (1 = exactly one, 0..1 = none or one, 1..N = one or many, 0..N = none, one or many; → = the foreign key points to)

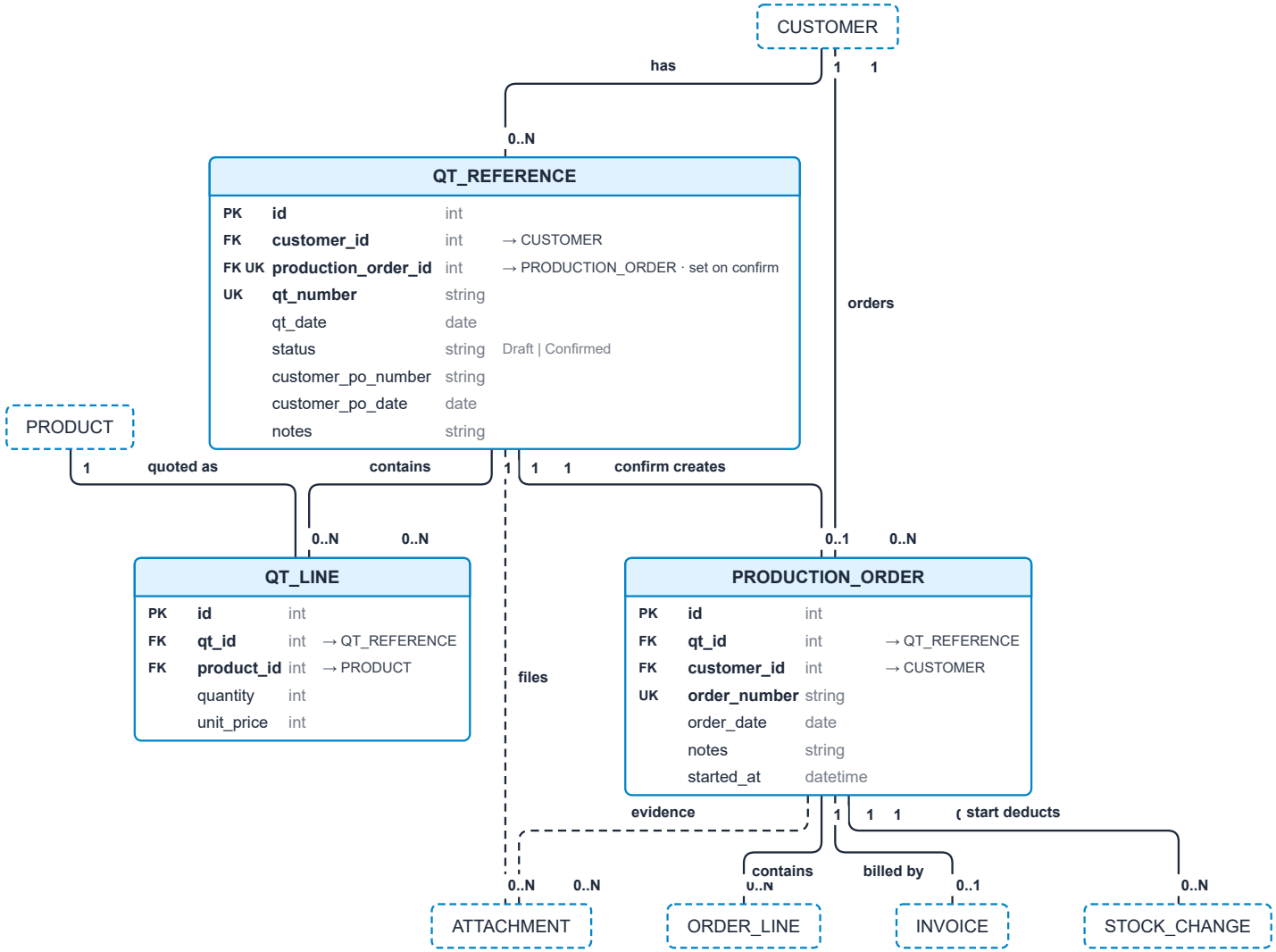



Figure 5. Sale: quotation (1 = exactly one, 0..1 = none or one, 1..N = one or many, 0..N = none, one or many; → = the foreign key points to)

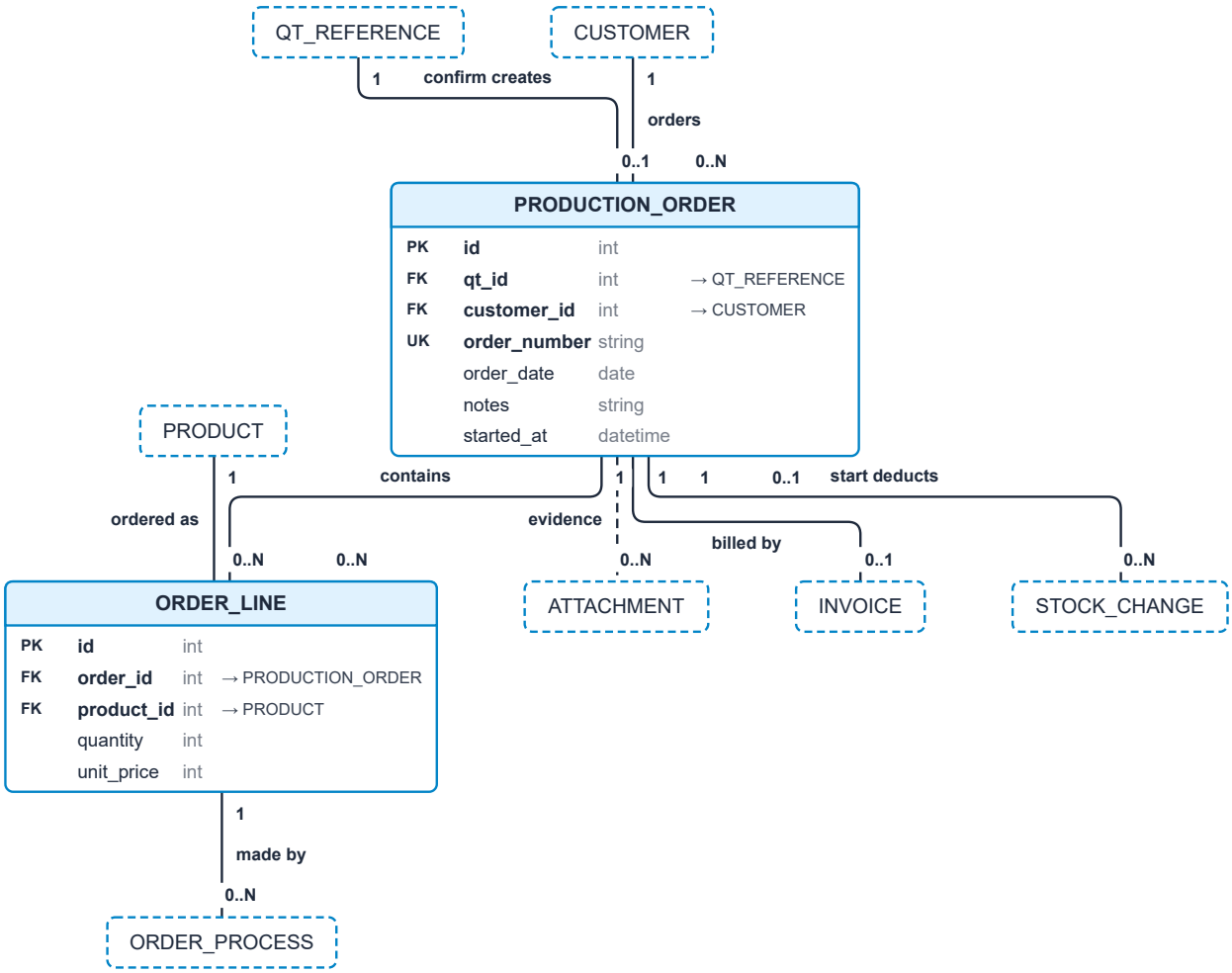



Figure 6. Sale: production order and its lines (1 = exactly one, 0..1 = none or one, 1..N = one or many, 0..N = none, one or many; → = the foreign key points to)

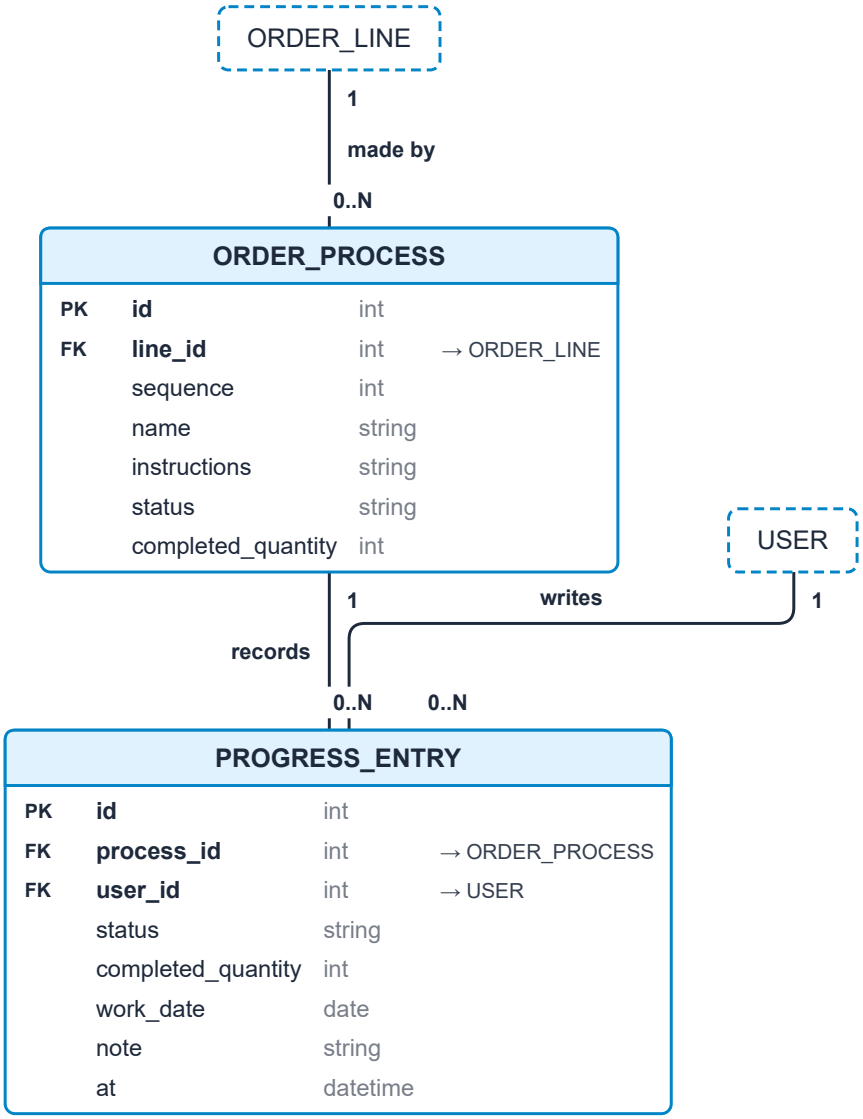



Figure 7. Sale: processes and progress (1 = exactly one, 0..1 = none or one, 1..N = one or many, 0..N = none, one or many; → = the foreign key points to)

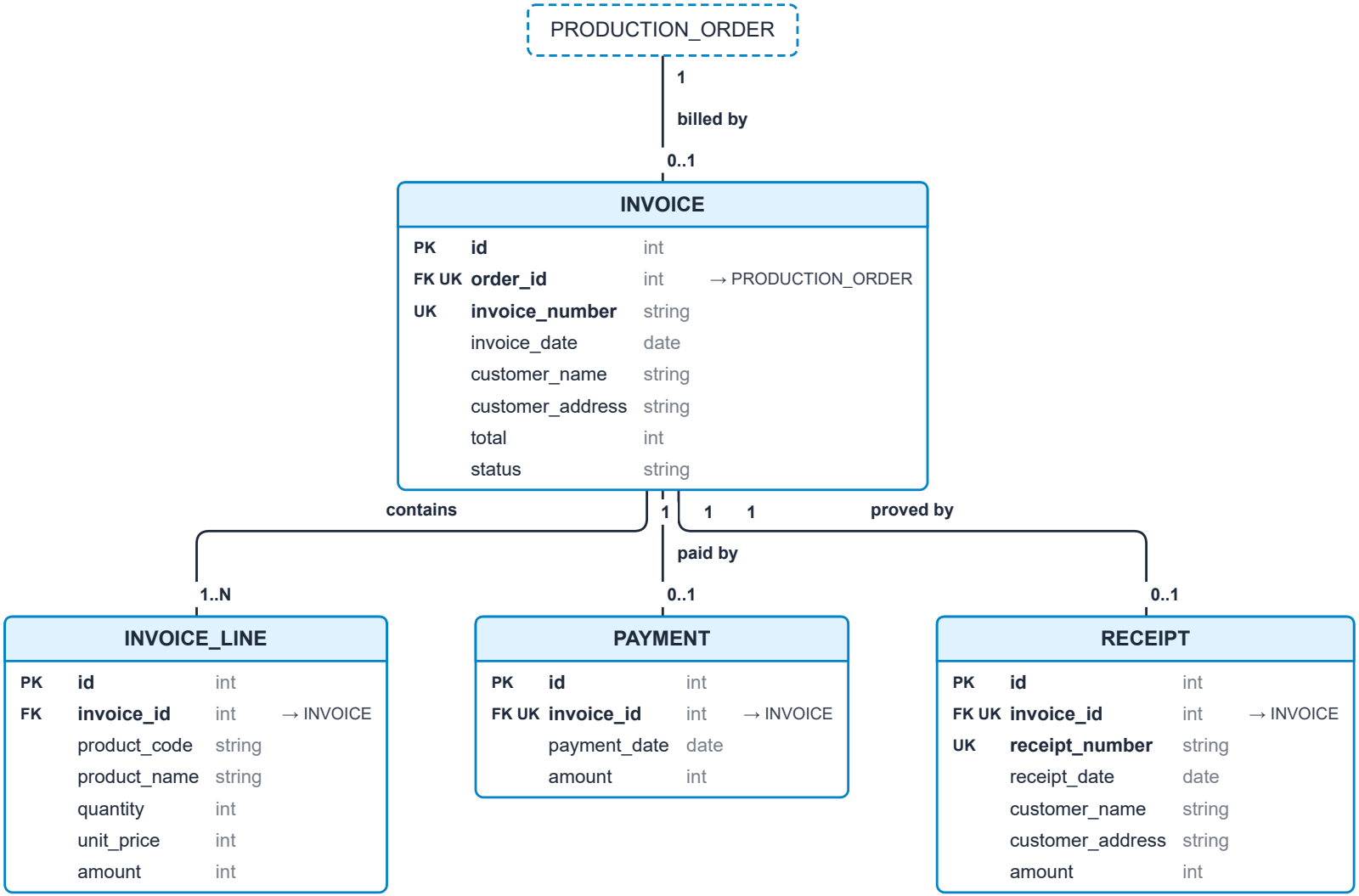



Figure 8. Accounting (1 = exactly one, 0..1 = none or one, 1..N = one or many, 0..N = none, one or many; → = the foreign key points to)

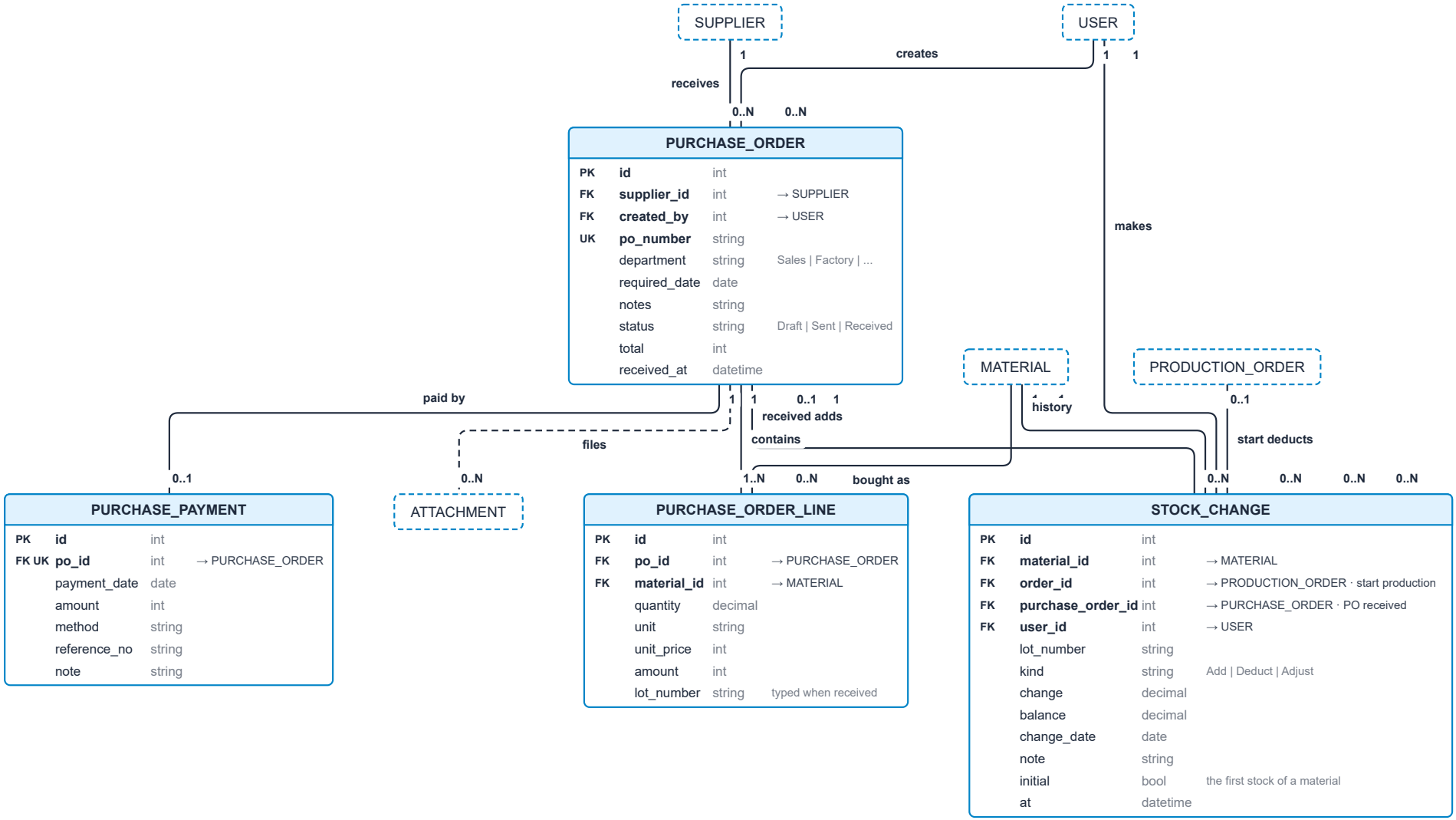



Figure 9. Purchase and stock (1 = exactly one, 0..1 = none or one, 1..N = one or many, 0..N = none, one or many; → = the foreign key points to)

## What Version 2.0 added

| Source | Change |
|---|---|
| Purchase | New tables `Supplier`, `SupplierMaterial`, `PurchaseOrder`, `PurchaseOrderLine`, `PurchasePayment`. `StockChange.purchase_order_id` links a received PO to the stock it added |
| QT 2.0 | `QtReference` gets `status`, `customer_po_number`, `customer_po_date`, `production_order_id`. New `QtLine`. The old QT file and the customer PO file move to `Attachment` |
| Files | New `Attachment` table, shared by PO, QT and production order (evidence). Replaces `file_name`, `file_type`, `file_path` on `QtReference` |
| User 2.0 | No new column. A Worker's `department` must be `Sales` or `Factory` and decides access |
| Material | `unit` added (the PO lines need a unit) |

## Notes

- `Attachment` is polymorphic (`owner_type` + `owner_id`), so there is no database foreign key to the owner; the service checks it.
- `PurchasePayment` is stored on the PO and does not go through the Accounting tables.
- `QT_REFERENCE.production_order_id` is unique: one QT creates one Factory Order. It is the second link between the two tables, in the other direction of "confirm creates" (`ProductionOrder.qt_id`).
- `QtLine` has no stored amount: `quantity x unit_price` is computed by the API. The process, progress and invoice line tables list the columns the code has.
- `Order status` (`Draft`, `In Progress`, `Completed`) is **computed** from the processes; it is not a column. The stock status (`Normal`, `Low`, `Out`) is computed from the lot totals and `min_level`.
- `Invoice` and `Receipt` copy the customer and the product data when they are issued (a snapshot). An order is billed once, and an invoice has at most one payment and one receipt.
- An order is made only by Confirm QT, so `ProductionOrder.qt_id` is always set and each QT has at most one order.
- A QT, an order and an order line may have no line or no process for a while (zero or many in the database): the first QT line is required when the QT is confirmed, and a line and a process on every line when the order starts. A purchase order and an invoice always have lines.
- `StockChange.initial` marks the first stock of a material; a material with a later change cannot be deleted.
- `ProductBom` and `SupplierMaterial` have a primary key of two columns (`PK,FK` on both).